① $F(s) = \dfrac{s^4 + 4s^2 + 3}{s(s^2+2)}$

a) $F(s) = \dfrac{k_0}{s} + k_\infty s + \sum_{i=1}^{\infty} \dfrac{2k_i s}{s^2 + \omega_i^2}$

$k_0 : \lim_{s\to 0} sF(s) = \lim_{s\to 0} s \cdot \dfrac{s^4+4s^2+3}{s(s^2+2)} = \dfrac{3}{2} = k_0$

$k_\infty : \lim_{s\to\infty} \dfrac{1}{s} F(s) = \lim_{s\to\infty} \dfrac{1}{s} \cdot \dfrac{s^4+4s^2+3}{s(s^2+2)} = \dfrac{s^4+4s^2+3}{s^4+2s^2}$

$k_\infty = 1$

$2k_i : \lim_{s^2\to -2} \dfrac{(s^2+2)}{s} \cdot F(s) = \lim_{s^2\to -2} \dfrac{(s^2+2)}{s} \dfrac{s^4+4s^2+3}{s(s^2+2)} = \dfrac{s^4+4s^2+3}{s^2}$

$2k_i = \dfrac{1}{2}$

$\Rightarrow F(s) = \dfrac{3/2}{s} + 1\,s + \dfrac{1/2\,s}{s^2+2}$

$F(s) = \dfrac{3}{2s} + s + \dfrac{1/2\,s}{s^2+2}$

$\dfrac{1/2\,s}{s^2+2} = \dfrac{1/C \cdot s}{s^2 + \frac{1}{LC}}$

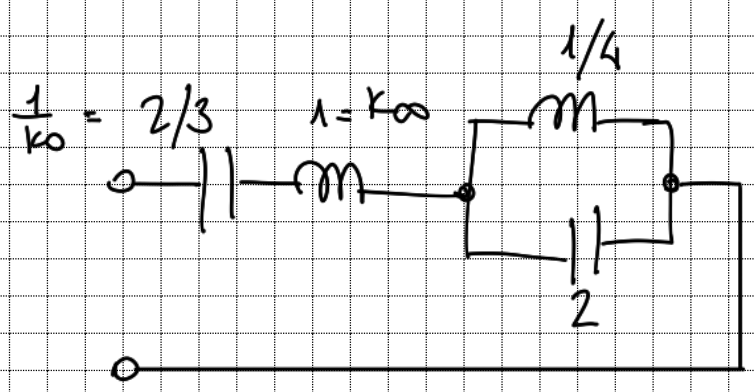
$\dfrac{1}{C} = \dfrac{1}{2}$

$C = 2$

$\Rightarrow \dfrac{1}{LC} = 2$

$\dfrac{1}{2L} = 2$

$L = \dfrac{1}{4}$



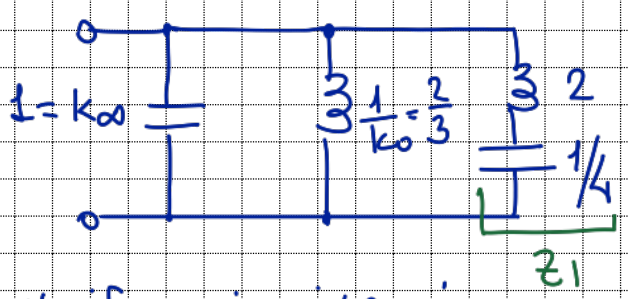
⑤ $F(S) = \frac{S^4 + 4S^2 + 3}{S(S+2)}$

$k_0 = \frac{3}{2}$ $\quad k_\infty = 1$ $\quad 2k_1 = \frac{1}{2}$ $\quad \frac{S \cdot 1/L}{S^2 + 1/LC}$

$\frac{1}{L} = \frac{1}{2} \Rightarrow L = 2$

$\frac{1}{LC} = 2 \Rightarrow C = \frac{1}{2L}$

$C = \frac{1}{4}$



verifico inmitancia

$z_1 = 2S + \frac{1}{\frac{1}{4}S} \Rightarrow 2S + \frac{4}{S} = \frac{2S^2 + 4}{S}$

$Y_1 = \frac{1}{z_1} = \frac{S}{2S^2+4} = \frac{S \cdot 1/2}{S^2+2}$

$Y(S) = Y_1 + \frac{1}{2S} + 1S \Rightarrow \frac{S\,1/2}{S^2+4} + \frac{3/2}{S} + S$

$Y(S) = \frac{1/2\,S^2 + 3/2\,(S^2+4) + S^2(S^2+2)}{S(S^2+4)}$

$Y(S) = \frac{1/2\,S^2 + 3/2\,S^2 + 6 + S^4 + 2S^2}{S(S^2+4)} = \frac{S^4 + 4S^2 + 6}{S(S^2+4)}$

(c) $F(s) = \frac{s^4 + 4s^2 + 3}{s(s^2+2)} = \frac{s^4+4s^2+3}{s^3+2s}$

$F(s) = Z(s) = \frac{K_0}{s} + K_\infty s + \sum \frac{2K_i s}{s^2+\omega^2} = \frac{s^4+4s^2+3}{s^3+2s}$

$K_\infty = 5$ (calculado anteriormente)

$$\frac{s^4+4s^2+3}{s^3+2s} - 5 = \frac{s^4+4s^2+3-s^4-2s^2}{s^3+2s}$$

$s^4+4s^2+3 \quad | \; s^3+2s$

$s^4+2s^2 \quad\quad\quad\quad s$

$2s^2+3$

$s^3+2s \quad | \; 2s^2+3$

$s^3+\frac{3}{2}s \quad\quad \frac{1}{2}s$

$\frac{1}{2}s$

$2s^2+3 \quad | \; \frac{1}{2}s$

$2s^2 \quad\quad 4s$

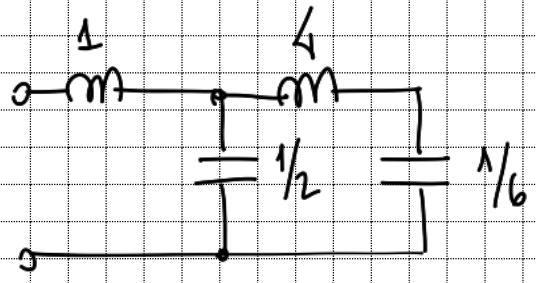
3

$\frac{1}{2}s \quad | \; 3$

$\frac{1}{2}s \quad\quad \frac{1}{6}s$

0

$\frac{2s^2+3}{s^3+2s}$ = $\frac{1}{1}$ + $\frac{1}{\frac{1}{2}s}$ + $\frac{1}{4s}$ + $\frac{1}{\frac{1}{6}s}$



(d)

$$Z_1(s) = \frac{s^4+4s^2+3}{s^3+2s} - \frac{3}{2s} = \frac{2s^5+8s^3+6s-3s^3-6s}{2s^4+4s^2}$$

$$\frac{2s^5+5s^3}{2s^4+4s^2} = \frac{2s^3(s^2+5/2)}{2s^2(s^2+2)} =$$

$$\frac{s(s^2+5/2)}{s^2+2}$$

$$Y_1(s) = \frac{1}{Z_1(s)} = \frac{s^2+2}{s(s^2+5/2)}$$

$$k_0' = \lim_{s\to 0} s \cdot \frac{s^2+2}{s(s^2+5/2)} = \frac{4}{5}$$

$$Y_2(s) = \frac{s^2+2}{s^3+\frac{5}{2}s} - \frac{4}{5}\frac{1}{s} = \frac{2s^2+4}{2s^3+5s} - \frac{4}{5}\cdot\frac{1}{s}$$

$$= \frac{10s^3+20s-8s^3-20s}{10s^4+25s^2} = \frac{2s^3}{10s^4+25s^2}$$

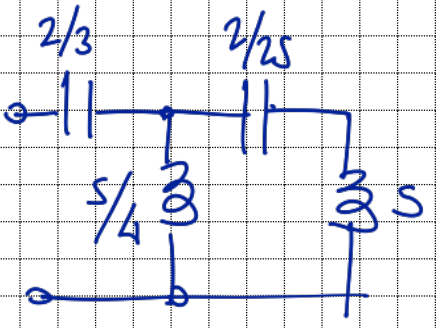
$$Y_2(s) = \frac{2}{10}\,\frac{s^3}{s^2}\,\frac{1}{s^2+\frac{25}{10}} \Rightarrow \boxed{Y_2(s) = \frac{1}{5}\,\frac{s}{s^2+\frac{5}{2}}}$$

$$Z_2(s) = 5\,\frac{(s^2+5/2)}{s}$$

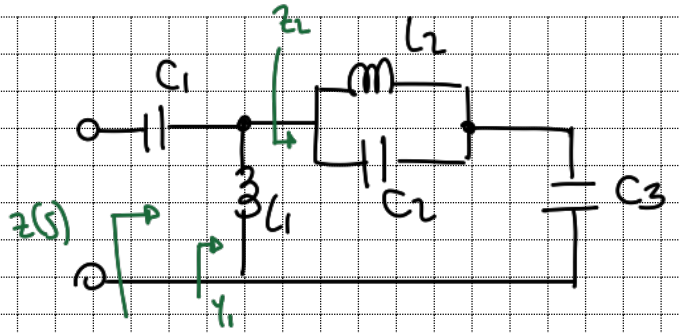
$$k_0'' = \lim_{s\to 0} s \cdot 5\,\frac{(s^2+5/2)}{s} = \frac{25}{2}$$

$$Z_3(s) = \frac{10s^2+25}{25s} - \frac{25}{25s} \qquad \boxed{Z_3(s) = 5s}$$

$$\Rightarrow \quad \frac{3}{2s} + \frac{1}{\frac{4}{5s}+\frac{1}{\frac{25}{25}+5s}}$$



(2)



$$Y(s) = \frac{s^5+18s^3+48s}{6s^4+42s^2+48}$$

$$Z(s) = \frac{1}{Y(s)}$$

$$Z(s) = 6\,\frac{s^4+7s^2+8}{s^5+18s^3+48s}$$

$$Z_1 = \lim_{s\to\infty} s\; 6\,\frac{s^4+7s^2+8}{s\,s^4+18s^2+48} = \frac{48}{48} = 1$$

$$\Rightarrow Z_1 = Z(s) - \frac{1}{s} \Rightarrow \frac{6s^4+42s^2+48}{s(s^4+18s^2+48)} - \frac{1}{s}$$

$$Z_1(s) = \frac{6s^4+42s^2+48}{s(s^4+18s^2+48)} - \frac{1}{s}\,\frac{s^4+18s^2+48}{s^4+18s^2+48}$$

$$Z_1(s) = \frac{6s^4-s^4+42s^2-18s^2+48-48}{s^5+18s^3+48s}$$

$$Z_1(s) = \frac{5s^4+24s^2}{s^5+18s^3+48s} = \frac{s^2(5s^2+24)}{s(s^4+18s^2+48)}$$

$$Z_1(s) = \frac{s(5s^2+24)}{s^4+18s^2+48}$$

$$Y_1(s) = \frac{s^4+18s^2+48}{s(5s^2+24)}$$

$$kd' = \lim_{s\to\infty} \frac{s^4+18s^2+48}{s(5s^2+24)} = \frac{48}{24} = 2 \qquad \frac{1}{L_1} = 2$$

$$\Rightarrow Y_2(s) = \frac{s^4+18s^2+48}{s(5s^2+24)} - \frac{2}{s}$$

$$Y_2(s) = \frac{s^4+18s^2+48-10s^2-48}{s(5s^2+24)} = \frac{s^2}{s}\,\frac{(s^2+8)}{(5s^2+24)} = s\,\frac{(s^2+8)}{5s^2+24}$$

$$Y_2(s) = \frac{s(s^2+8)}{5s^2+24} \qquad Z_2(s) = \frac{5s^2+24}{s(s^2+8)}$$

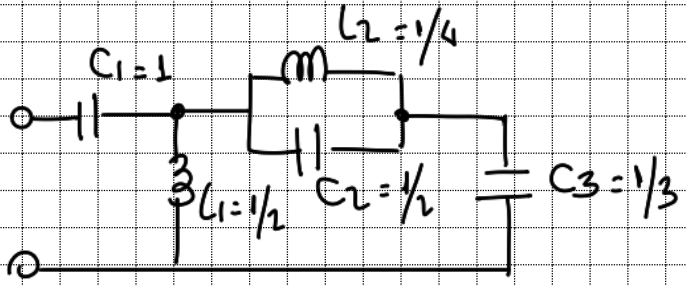
Removo C_3

$$k_0^{\infty} = \lim_{s\to 0} s\,\frac{5s^2+24}{s(s^2+8)} = 3 \qquad C_3 = 1/3$$

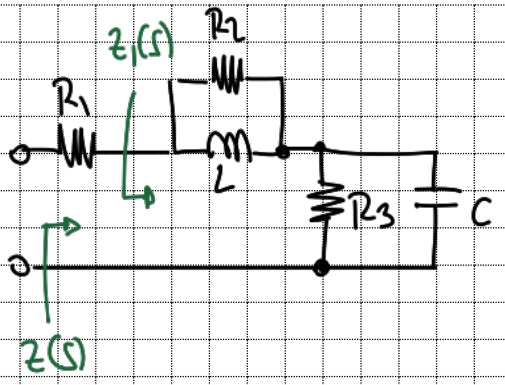
$$Z_3(s) = \frac{5s^2+24}{s(s^2+8)} - \frac{3}{s} = \frac{5s^2+24-3s^2-24}{s(s^2+8)} = \frac{2s^2}{s(s^2+8)}$$

$$Z_3(s) = \frac{2s}{s^2+8}$$

$$Y_3(s) = \frac{s^2+8}{2s} = \frac{s}{2} + \frac{4}{s} \qquad C_2 = 1/2 \qquad L_2 = 1/4$$



③



$$z(s) = \frac{s^2+6s+8}{s^2+6s+5} = R_1 + z_{RL}(s) + z_{RC}(s)$$

$$z(s) = \frac{s^2+6s+8}{s^2+6s+5} = \frac{(s+2)(s+4)}{(s+1)(s+5)}$$

$$K_\infty = \lim_{s\to\infty} \frac{s^2+6s+8}{s^2+6s+5} = 1 = R_1 + R_2$$

$$K_0 = \lim_{s\to 0} \frac{s^2+6s+8}{s^2+6s+5} = \frac{8}{5} = R_1 + R_3$$

$$\lim_{s\to -1} (s+1)\frac{(s+4)(s+2)}{(s+1)(s+5)} = \frac{3}{4} \qquad z_{RL} = \frac{R_2 s}{s + R_2/L}$$

$$\lim_{s\to -5} (s+5)\frac{(s+4)(s+2)}{(s+1)(s+5)} = -\frac{3}{4} \qquad z_{RC} = \frac{1/C}{s + 1/R_3 C}$$

$$1 + \frac{3}{4(s+1)} - \frac{3}{4(s+5)} = R_1 + \frac{R_2 s}{s + R_2/L} + \frac{1/C}{s + 1/R_3 C}$$

$$\frac{4(s+1)(s+5) + 3(s+5) - 3(s+1)}{4(s+1)(s+5)}$$

$$\frac{4(s^2+6s+5) + 3s + 15 - 3s - 3}{4(s^2+6s+5)}$$

$$\frac{4(s^2+6s+5) + 4(3)}{4(s^2+6s+5)} = \frac{s^2+6s+8}{s^2+6s+5}$$

$\Rightarrow$

$$1 + \frac{3}{4(s+1)} - \frac{3}{4(s+1)} = R_1 + \frac{R_2 s}{s + R_2/L} + \frac{1/C}{s + 1/R_3 C}$$

$$\frac{3/4}{s+1} = \frac{1/C}{s + 1/R_3 C} \qquad \begin{cases} \frac{1}{C} = \frac{3}{4} \Rightarrow C = \frac{4}{3} \\ \frac{1}{R_3 C} = 1 \Rightarrow R_3 = \frac{1}{C} \Rightarrow R_3 = \frac{3}{4} \end{cases}$$

$$\begin{cases} R_1 + R_2 = 1 \\ R_1 + R_3 = 8/5 \end{cases} \quad \wedge \; R_3 = \frac{3}{4} \Rightarrow R_1 = \frac{8}{5} - \frac{3}{4}$$

$$\frac{17}{20} + R_2 = 1 \qquad R_1 = \frac{32}{20} - \frac{15}{20} \Rightarrow R_1 = \frac{17}{20}$$

$$\Rightarrow R_2 = 1 - \frac{17}{20} = \frac{3}{20}$$

$$\frac{-3/4}{s+5} = \frac{s R_2}{s + R_2/L} \qquad \frac{R_2}{L} = 5 \Rightarrow L = \frac{R_2}{5}$$

$$L = \frac{3/20}{5} \Rightarrow L = \frac{3}{100}$$

